

Yes — this **Week-5 BA task** is a **Python-based financial analytics project**, not a SQL project.

The project is:

Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

Main objective

You need to build a simulator that:

1. Works with **500 assets**
2. Calculates **historical returns**
3. Calculates the **500 × 500 covariance matrix**
4. Performs **Cholesky decomposition**
5. Generates correlated random numbers
6. Simulates future asset prices using **Geometric Brownian Motion (GBM)**
7. Calculates future portfolio values
8. Calculates **99% Value-at-Risk (VaR)**
9. Identifies portfolio risk

Overall workflow

```
Historical Price Data
    ↓
Calculate Returns
    ↓
500 × 500 Covariance Matrix
    ↓
Cholesky Decomposition
    ↓
Generate Random Noise
    ↓
Correlate Random Noise
    ↓
Geometric Brownian Motion
    ↓
Simulate Portfolio Prices
    ↓
Calculate Portfolio Values
    ↓
Calculate Profit / Loss
    ↓
99% Value-at-Risk (VaR)
    ↓
Risk Analysis & Graphs
```

Important formulas

Historical return:

$$R_t = P_t - P_{t-1} \quad R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Covariance matrix:

$$\Sigma = \text{Cov}(R) \quad \Sigma = \text{Cov}(R)$$

For 500 assets, the covariance matrix will contain:

$$500 \times 500 = 250,000 \quad 500 \times 500 = 250,000$$

covariance values.

Cholesky decomposition:

$$\Sigma = LL^T \quad \Sigma = LL^T$$

where L is the lower-triangular Cholesky matrix.

Geometric Brownian Motion:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\Delta t Z\right] \quad S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z\right]$$

99% VaR:

If your simulated portfolio losses are sorted, the 99% VaR corresponds approximately to the **1st percentile of portfolio returns/losses**.

Python libraries you will typically use

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

For a 500-asset project, **NumPy** is especially important because matrix operations and simulation need to be efficient.

If you have a **dataset for this BA task**, upload it and I can build the complete project **step-by-step from preprocessing → covariance matrix → Cholesky → GBM → Monte Carlo simulation → 99% VaR → graphs**, with beginner-level explanation of every line.